

Question 1

A circle has center A , and points B and C lie on the circle. Line segments BD and CD are tangent to the circle at points B and C , respectively. The measure of angle BDC is 120° . The length of line segment CD is 12 inches. What is the area of the region that is both inside quadrilateral $ABDC$ and outside the circle, in square inches? (Round your answer to the nearest hundredth of a square inch)

Question 2

$$23 - 7x = \sqrt{3k - 2x}$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of $56k$?

Question 3

For the function f , for each increase in the value of x by $\frac{1}{4}$, the value of $f(x)$ increases by a factor of m . Which of the following equivalent forms displays m as a constant or coefficient?

A) $f(x) = 13(100)^{2x}$

B) $f(x) = 13(1000^{\frac{2}{7}})^{\frac{14}{3}x}$

C) $f(x) = 13(10^{\frac{2}{3}})^{6x}$

D) $f(x) = 13(100000^{\frac{16}{5}})^{\frac{1}{4}x}$

Question 4

A juice mixture has a total volume of 125 liters and is 45% juice by volume. The mixture was created by mixing two different solutions (solution A and solution B).

Solution A was 24% juice by volume. Solution B was 64% juice by volume.

The volume, in liters, of juice in solution A is equal to a .

The volume, in liters, of juice in solution B is equal to b .

The numbers a , b , and p are all positive constants.

If a is $p\%$ less than b , what is the value of p ? (Round your answer to the nearest hundredth)

Question 5

In triangle DEF , angle F is a right angle, point Q lies on $\overline{DE}$, point J lies on $\overline{EF}$, and $\overline{QJ}$ is parallel to $\overline{DF}$. If the length of $\overline{EF}$ is 45 units, the length of $\overline{FJ}$ is 24 units, and the area of triangle DEF is 630 square units, what is the perimeter of triangle $Q EJ$, in units?

Question 6

The equation below is true for all values of x where $x \neq \frac{5}{a}$ and $x \neq -\frac{3b}{2}$. The numbers a and b are constants. What is the value of $\frac{b}{a}$?

$$\frac{3x}{ax - 5} - \frac{7}{2x + 3b} = \frac{6x^2 + 5x + a + b}{(ax - 5)(2x + 3b)}$$

Question 7

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For what fraction of the data points in the scatterplot is the y -value predicted by the line of best fit less than the actual y -value?

